

VAUXHALL, AB, Canada

WMO: 712510

Lat: 50.053N

Lon: 112.126W

Elev: 2556

StdP: 13.39

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-15.9	-28.3	1.2	-22.0	-21.5	1.8	-15.8	35.0	40.5	31.5	37.6	4.4	210	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.6	89.4	64.3	85.9	63.2	82.6	62.2	68.1	82.1	66.1	80.0	64.2	78.0	9.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	97.1	73.6	61.2	88.9	71.6	59.1	82.3	69.4	34.0	81.8	32.3	79.9	30.7	78.0	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	25.6	22.2	DB	-31.0	95.6	7.3	3.1	-36.3	97.9	-40.5	99.7	-44.6	101.5	-49.9	103.7	(4)
(5)			WB	-31.2	72.1	7.2	2.9	-36.4	74.2	-40.7	76.0	-44.7	77.6	-50.0	79.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	17.4	20.5	30.5	42.6	52.7	60.0	66.2	64.5	55.7	43.2	29.6	19.3	(6)
(7)		DBStd	20.75	17.09	15.95	14.69	8.95	7.98	5.68	5.22	5.81	7.90	9.16	13.99	15.64	(7)
(8)		HDD50	4605	1011	825	607	248	64	3	0	1	34	244	615	953	(8)
(9)		HDD65	8597	1476	1245	1070	672	387	169	48	81	292	678	1061	1418	(9)
(10)		CDD50	1669	0	0	2	26	147	302	502	450	204	32	4	0	(10)
(11)		CDD65	186	0	0	0	0	5	18	85	66	12	0	0	0	(11)
(12)		CDH74	3721	0	0	1	21	210	369	1409	1286	410	15	0	0	(12)
(13)		CDH80	1272	0	0	0	2	49	83	509	494	134	1	0	0	(13)
(14)	Wind	WSAvg	9.1	9.6	8.9	10.0	10.4	9.7	8.6	7.4	7.1	8.1	9.8	9.9	9.7	(14)
(15)	Precipitation	PrecAvg	13.8	0.7	0.5	0.8	1.3	1.6	2.4	1.4	1.5	1.5	0.6	0.6	0.7	(15)
(16)		PrecMax	19.3	1.7	1.1	1.6	4.4	5.2	5.9	3.1	3.9	6.5	1.7	1.7	2.7	(16)
(17)		PrecMin	9.8	0.0	0.0	0.1	0.0	0.2	0.4	0.1	0.2	0.2	0.0	0.1	0.1	(17)
(18)		PrecStd	3.3	0.5	0.3	0.5	1.1	1.1	1.5	0.9	0.9	1.4	0.5	0.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	56.7	64.7	77.0	85.4	87.6	93.4	93.8	89.7	76.0	64.1	53.5	(19)
(20)			MCWB	44.4	43.5	47.7	53.0	58.9	66.3	66.0	63.8	60.1	55.3	49.7	42.9	(20)
(21)		2%	DB	47.4	49.9	59.4	70.3	80.0	82.2	89.4	89.4	83.8	69.3	57.3	47.9	(21)
(22)			MCWB	40.1	39.8	45.8	50.9	57.0	62.1	66.7	63.0	59.1	52.6	46.0	39.6	(22)
(23)		5%	DB	43.8	44.9	54.8	64.8	74.7	78.3	85.7	85.8	78.7	64.0	52.8	43.9	(23)
(24)			MCWB	38.0	37.2	43.2	47.8	55.0	60.4	65.4	62.8	57.7	50.2	43.4	37.2	(24)
(25)		10%	DB	40.1	40.8	49.7	59.6	70.1	74.6	82.1	81.7	73.3	59.2	48.1	40.4	(25)
(26)			MCWB	35.6	35.1	40.4	45.5	52.9	58.7	64.3	62.0	55.9	47.3	40.2	34.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	44.7	49.8	55.0	61.7	68.6	73.0	68.9	63.5	57.3	51.2	44.2	(27)
(28)			MADB	51.2	55.7	62.1	72.4	79.3	82.0	85.1	82.2	82.1	72.3	61.9	51.9	(28)
(29)		2%	WB	40.8	40.5	46.6	51.7	58.7	65.0	69.3	66.5	60.7	53.5	46.8	40.3	(29)
(30)			MADB	46.9	48.2	57.5	67.7	74.2	77.5	82.8	81.3	78.9	67.4	56.1	46.6	(30)
(31)		5%	WB	38.0	37.8	43.7	49.1	56.4	62.5	67.3	64.7	58.6	50.6	43.7	37.7	(31)
(32)			MADB	43.1	44.2	53.5	62.9	71.9	74.1	81.2	80.0	75.4	62.5	52.1	43.4	(32)
(33)		10%	WB	35.7	35.2	40.8	46.4	54.2	60.2	65.4	62.9	56.4	48.1	40.7	35.1	(33)
(34)			MADB	40.0	40.4	49.2	58.0	68.1	71.5	78.8	78.3	71.4	58.0	47.3	40.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.2	21.6	22.1	24.0	24.7	23.1	27.6	28.7	27.2	24.1	20.9	20.0	(35)
(36)			MCDBR	20.9	25.1	29.0	34.8	33.7	30.8	34.5	37.5	38.7	32.3	25.8	21.1	(36)
(37)			MCWBR	15.1	16.7	17.1	17.7	14.8	14.2	15.0	15.0	16.9	17.4	15.9	14.7	(37)
(38)		5% WB	MCDBR	20.6	24.4	27.9	31.6	29.6	27.5	31.1	31.7	34.9	30.8	24.9	21.3	(38)
(39)			MCWBR	15.5	17.3	17.2	17.0	13.8	13.9	15.2	14.0	16.2	17.2	16.1	15.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.264	0.272	0.296	0.338	0.354	0.349	0.350	0.370	0.327	0.293	0.266	0.251	(40)
(41)		taud		2.438	2.472	2.455	2.343	2.345	2.410	2.421	2.343	2.441	2.512	2.489	2.431	(41)
(42)		Ebn at Noon		254	282	292	288	286	288	286	274	276	269	252	243	(42)
(43)		Edh at Noon		19	23	29	36	37	36	35	36	29	23	18	16	(43)
(44)	All-Sky Solar Radiation	RadAvg		390	705	1120	1546	1840	2020	2206	1765	1256	778	431	302	(44)
(45)		RadStd		34	60	84	107	137	131	111	111	108	82	35	27	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page